TASK-005: Regression and Security Test Cases

Project: Student Attendance Management System **Task:** Regression and Security Testing

Test Cases

Test ID	Test Scenario	Expected Result	Recorded Result
REG-001	Admin login with valid credentials	Login succeeds	Pass — HTTP 200
REG-002	Admin login with invalid password	Login rejected	Pass — HTTP 401
REG-003	Create a class	Class created successfully	Pass — HTTP 200
REG-004	Create a student	Student created successfully	Pass — HTTP 200
REG-005	Create student with invalid email	Request rejected	Pass — HTTP 400
REG-006	Create student with duplicate email	Duplicate rejected	Pass — HTTP 409
REG-007	Create a subject	Subject created successfully	Pass on retry — HTTP 200; initial HTTP 500
REG-008	Create faculty	Faculty created successfully	Pass — HTTP 200
REG-009	Assign faculty	Assignment succeeds	Pass — HTTP 200
REG-010	Mark attendance	Attendance recorded	Pass — HTTP 200
REG-011	Submit duplicate attendance	Duplicate rejected	Pass — HTTP 409
REG-012	Submit invalid attendance status	Request rejected	Pass — HTTP 400
REG-013	Request unknown session	Not found response	Pass — HTTP 404
REG-014	Attendance for a student not enrolled	Request rejected	Pass — HTTP 400
REG-015	View attendance history	History displayed	Pass — HTTP 200
REG-016	View attendance percentage	Percentage returned	Pass — HTTP 200
REG-017	View low-attendance list	List returned	Pass — HTTP 200
REG-018	Generate attendance report	Report returned	Pass — HTTP 200
REG-019	Export attendance as CSV	CSV response returned	Pass — HTTP 200
REG-020	Correct attendance record	Correction succeeds and audit entry recorded	Pass — HTTP 200
SEC-001	Student accesses admin-only audit endpoint	Access denied	Pass — HTTP 403
SEC-002	Reuse token after logout	Access rejected	Pass — HTTP 401
SEC-003	Access protected endpoint without authentication	Access rejected	Pass — HTTP 401
SEC-004	SQL injection-style student search input	Input handled safely	Smoke test passed — HTTP 200, empty result

Test ID	Test Scenario	Expected Result	Recorded Result
SEC-005	Verify password storage	Salted PBKDF2 hashes used	Code inspection recorded
SEC-006	Verify session expiry	Expired session rejected	Pass — HTTP 401 after configured 5-second timeout
SEC-007	Check HTTPS support	HTTPS connection succeeds	Pending — HTTPS not configured locally
REG-021	Run automated tests	Tests execute successfully	2 passed; 18 warnings
REG-022	Basic concurrent requests	Requests return successfully	Smoke test passed — 5/5 HTTP 200

Notes and Limitations

• Results reflect recorded observations and do not establish complete security assurance.
• Subject creation initially returned HTTP 500 due to a transient SQLite lock; retry returned HTTP 200.
• HTTPS deployment testing remains pending because HTTPS was not configured locally.
• SQL injection and concurrent-request checks were limited smoke tests.
• Further browser, mobile, and deployment testing may be required.

Conclusion

Recorded tests cover core attendance workflows, authentication, authorization, validation, reporting, and selected security controls. Outstanding limitations should be addressed before final release approval.